World History: Gutenberg Printing Press - Page 1

Unit 4: The Information Revolution in Europe

Around 1440 in Mainz, Germany, goldsmith Johannes Gutenberg invented the movable metal type printing press. Before Gutenberg, every book in Europe had to be painstakingly copied by hand by scribes or block-printed on wood.

Gutenberg combined three breakthrough innovations:

1. Durable movable type cast from an alloy of lead, tin, and antimony.
2. Specially formulated oil-based ink that adhered firmly to metal surfaces.
3. A heavy wooden screw press adapted from agricultural wine presses.

The Gutenberg Bible of 1455 proved that mechanically printed books could match the beauty of illuminated manuscripts while producing dozens of copies at once.

World History: Gutenberg Printing Press - Page 2

Social and Cultural Impact

Within 50 years of Gutenberg’s invention, print shops operated in over 250 European cities, producing an estimated 20 million volumes by 1500.

Major consequences included:

- Book costs plummeted by over 80 percent, dramatically expanding public literacy.
- Scientific findings could be replicated, cross-checked, and distributed across borders.
- Martin Luther’s 95 Theses spread across Europe in weeks, accelerating the Reformation.